

## **(Optional) Git Authentication on the Command Line**

Use this only if you want to push from the terminal instead of VS Code.

# Learning Goals

- Explain why GitHub rejects your account password in Git commands.
- Sign in with GitHub CLI, the simplest command-line method.
- Use a personal access token if GitHub CLI does not work.

# Why a Terminal Push Fails

- A **public** repository can be cloned without signing in.
- `git push` always needs you to be signed in.

If you push from the terminal and type your account password, you see:

```
remote: Invalid username or token. Password authentication is not supported for Git operations.  
fatal: Authentication failed for 'https://github.com/yourusername/yourusername.github.io.git/'
```

GitHub does not accept your account password for Git commands over HTTPS.

- In this course, VS Code signs you in through the browser, so you do not need this page.
- Read on only if you want to use the terminal.

## Option 1: GitHub CLI (Simplest)

Sign in once with one command.

- Install it first: `brew install gh` on macOS.
- On Ubuntu / WSL2, follow [github.com/cli/cli#installation](https://github.com/cli/cli#installation).

Then run:

```
gh auth login
```

| Prompt                                         | Choose                   |
|------------------------------------------------|--------------------------|
| Account to log into                            | GitHub.com               |
| Preferred protocol for Git operations          | HTTPS                    |
| Authenticate Git with your GitHub credentials? | Y                        |
| How to authenticate                            | Login with a web browser |

Enter the one-time code in the browser. Then run `git push` again.

## Option 2: Personal Access Token (Last Resort)

A **personal access token** is a password-like string that replaces your account password.

- Create your `yourusername.github.io` repository first.

1. On GitHub: profile picture → **Settings** → **Developer settings**.
2. **Personal access tokens** → **Fine-grained tokens** → **Generate new token**.
3. Choose a name and an **Expiration**.
4. **Repository access: Only select repositories** → `yourusername.github.io`.
5. **Repository permissions: Contents** → **Read and write**.
6. Click **Generate token** and copy it. GitHub shows it only once.

# Use the Token

When Git asks, type your username and paste the token as the password:

```
Username for 'https://github.com': yourusername  
Password for 'https://yourusername@github.com': <paste your token>
```

- Nothing appears while you paste. Press Enter.
- If it still fails: the token expired, **Contents** is not **Read and write**, or an old password is saved in your Keychain or credential manager.

Treat a token like a password. Never commit it or share it.

If it leaks, delete it in **Settings** → **Developer settings**, and create a new one.